

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Paper
reference

8FM0/22

Further Mathematics

Advanced Subsidiary

Further Mathematics options

22: Further Pure Mathematics 2

(Part of option A only)

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
- Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Q:1/1/1/1/




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1. Sketch on an Argand diagram the region defined by

$$\left\{z \in \mathbb{C} : -\frac{\pi}{4} < \arg(z + 2) < \frac{\pi}{4}\right\} \cap \{z \in \mathbb{C} : -1 < \operatorname{Re}(z) \leq 1\}$$

On your sketch

- shade the part of the diagram that is included in the region
- use solid lines to show the parts of the boundary that are included in the region
- use dashed lines to show the parts of the boundary that are not included in the region

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$\mathbf{M} = \begin{pmatrix} 4 & 2 \\ 3 & -1 \end{pmatrix}$$

Find a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{P}^{-1}\mathbf{M}\mathbf{P} = \mathbf{D}$$

(7)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



- (3)

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|---------------|----|----|----|----|----|----|----|----|
| $\times_{30}$ | 2 | 4 | 8 | 14 | 16 | 22 | 26 | 28 |
| 2 | 4 | 8 | 16 | 28 | 2 | 14 | 22 | 26 |
| 4 | 8 | | 2 | | | 28 | 14 | |
| 8 | 16 | 2 | | | 8 | | | 14 |
| 14 | 28 | | 22 | 16 | | 8 | 4 | |
| 16 | 2 | 4 | | 14 | 16 | | | |
| 22 | 14 | | 26 | | | 4 | 2 | 16 |
| 26 | 22 | 14 | | 4 | | | | 8 |
| 28 | 26 | | 14 | | 28 | | 8 | |

(b) Hence determine whether the set X with the operation of multiplication modulo 30 forms a group.

(6)

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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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$\times_{30}$	2	4	8	14	16	22	26	28
2	4	8	16	28	2	14	22	26
4	8		2			28	14	
8	16	2			8			14
14	28		22	16		8	4	
16	2	4		14	16			
22	14		26			4	2	16
26	22	14		4				8
28	26		14		28		8	



4.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (i) (a) Use the Euclidean algorithm to find the highest common factor h of 416 and 72

- (b) Hence determine integers a and b such that

$$416a + 72b = h \tag{3}$$

- (c) Determine the value c in the set $\{0,1,2,\dots,415\}$ such that

$$23 \times 72 \equiv c \pmod{416} \tag{2}$$

- (ii) Evaluate $5^{10} \pmod{13}$ giving your answer as the smallest positive integer solution. **(3)**



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Question 4 continued

Handwriting practice area with horizontal lines.



Question 4 continued

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 11 marks)



5. A person takes a course of a particular vitamin.

Before the course there was none of the vitamin in the person's body.

During the course, vitamin tablets are taken at the same time each day.

Initially two tablets are taken and on each following day only one tablet is taken.

Each tablet contains 10 mg of the vitamin.

Between doses the amount of the vitamin in the person's body decreases naturally by 60%

Let u_n mg be the amount of the vitamin in the person's body immediately after a tablet is taken, n days after the initial two tablets were taken.

- (a) Explain why u_n satisfies the recurrence relation

$$u_0 = 20 \quad u_{n+1} = 0.4u_n + 10 \quad (2)$$

The general solution to this recurrence relation has the form $u_n = a(0.4)^n + b$

- (b) Determine the value of a and the value of b . (4)

The course is only effective if the amount of the vitamin in the person's body remains above 6 mg at **all times** throughout the course.

- (c) Determine whether this course of the vitamin will be effective for this person, giving a reason for your answer. (3)



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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 9 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS

